Amr Belal Arab

AI Engineer

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My-portfolio

Summary

My journey in AI began with building smart solutions that matter — from developing a chatbot for mental health support using LSTM + Attention, to deploying a secure chat interface that prevents data leakage in organizations. I'm driven by the impact of real-time, intelligent systems. Whether it's visualizing urban taxi trends or clustering customer behavior, I enjoy turning data into action. I thrive at the intersection of AI, software, and real-world problems.

Education

Helwan International Technological University (HITU) **2022-2026**

- **Artificial intelligence Department**
- **GPA: 3.78**
- **Completed coursework:** Applied Deep Learning
- **Completed coursework:** Python for AI.
- **Completed coursework:** Python.

Experience

Freelance AI Developer **Mar 2024 - Present**

- Designed and implemented AI-based solutions for various clients in different domains.
- Applied techniques such as outlier detection, clustering, and NLP for real-world datasets.

Projects

Weekly GenAI LinkedIn Agent – private project **Jul-2026**

- Built a private multi-agent system that researches current Generative AI news, verifies stories across multiple sources, and ranks the strongest updates.
- Generated professional Arabic and English LinkedIn posts, applied an automated quality gate, and delivered deduplicated email digests on schedule.

Technologies: Python, Agentic AI, Multi-Agent Systems, Gemini API, Structured Output, Automated Quality Gates, SMTP, GitHub Actions, Pytest.

ISH - Intelligent Spherical Home Robot **2025 - 2026 - GitHub-link | Documentation**

- Developed ISH, a BB-8-inspired AI-powered spherical smart-home robot and **college graduation project** that unifies robotics, computer vision, speech AI, environmental sensing, a FastAPI backend, and a Flutter mobile application.
- Designed it as a mobile household assistant, safety monitor, and security platform that helps residents identify people, check home conditions, receive alerts, manage reminders, remember object locations, and access the robot remotely.
- Key features include face recognition and authorization, live ESP32-CAM streaming, Arabic/English voice commands with local Egyptian Arabic replies, environmental risk monitoring, reminders, object-location memory, mobile access, and Supabase-backed event history.

Technologies: Python, FastAPI, Computer Vision, InsightFace, Whisper STT, Local TTS, Flutter, Supabase, ESP32, Raspberry Pi, IoT, Robotics.

Raay - Amazon Egypt Sentiment & AI Marketing **Jul-2026 - GitHub-link | Live Demo**

- Built the **DEPI Round 4 GenAI-track graduation project** to search Amazon Egypt, collect Arabic and English reviews, and analyze product sentiment.
- Compared and ranked products, visualized insights, exported results, and generated a social-ready AI marketing poster for the strongest product.

Technologies: Python, Streamlit, Arabic NLP, Sentiment Analysis, Web Scraping, Transformers, HuggingFace, Stable Diffusion, Pillow, Data Visualization, Pytest.

SourceCraft - Multi-Agent Research Assistant Jul-2026 - GitHub-link | Live Demo

- Built a source-aware three-agent research assistant that searches the web and writes cited Markdown briefs using a bounded LangGraph workflow.
- Added automated review and evidence-aware routing that sends writing issues to the Writer and evidence gaps back to the Researcher.

Technologies: Python, LangGraph, Gemini API, Multi-Agent Systems, DuckDuckGo Search, Streamlit, Structured State, Pytest, Ruff, Docker, CI.

Course Registration System - HCI Prototype May-2026 - GitHub-link

- Created a three-tier course-registration HCI prototype with administrator authentication, dashboards, CRUD workflows, advanced search, and validation.
- Applied HCI principles including visibility, consistency, feedback, and error prevention across the interface.

Technologies: Python, Flask, SQLite, HTML, CSS, JavaScript, Bootstrap 5, 3-Tier Architecture, HCI, CRUD.

FireVision Classifier – Fire Detection Apr-2026 – GitHub-link

- Developed a computer vision classification system for fire vs non-fire image recognition using EfficientNetB0 and transfer learning.
- Designed and evaluated 88 preprocessing experiments, including augmentation, edge detection, and hybrid pipelines, to identify the most effective strategy for fire image classification.
- Improved model performance to 97.96% accuracy, with augmentation-based preprocessing outperforming edge-detection-based methods.

Technologies: Python, TensorFlow, Keras, EfficientNetB0, OpenCV, Albumentations, Transfer Learning, Deep Learning, Computer Vision, Image Classification.

IOT Weather Prediction & Smart Control System Nov-2025 – GitHub-link

- Designed a smart IoT system to monitor temperature, humidity, rain, wind, and light using ESP32 and real sensors.
- Trained an ML prediction model to generate “Play/Not Play” decisions based on real-time sensor data.
- Created a real-time dashboard to visualize all readings and toggle between Manual and AI control modes.

Technologies: ESP32, Sensors, PWM RGB Control, Python, ML (Naïve Bayes), php, MySQL Database, Plotly.js, HTML/CSS/JavaScript, IoT Data Pipeline.

Information Retrieval Engine & Flask web Interface Nov-2025 – GitHub-link

- Built a complete Information Retrieval pipeline using NLTK preprocessing and Gensim’s TF-IDF model to retrieve, rank, and evaluate over 2,000 movie review documents (POS/NEG).
- Deployed a Flask-based search interface displaying top-k results with automated precision, recall, and F1-score evaluation.

Technologies: Python, NLTK, Gensim, Scikit-learn, Flask, TF-IDF, Cosine Similarity, Information Retrieval, Web Deployment.

Drug Classification – Evaluation of ML Models & Scaling Techniques Jul-2025 – GitHub-link

- Built a complete ML pipeline to classify drug types using Decision Tree, SVM, KNN, and Logistic Regression with multiple scaling techniques; evaluated models using accuracy, precision, recall, and F1-score.
- Applied feature encoding, scaling (MinMax, Standard, Robust, MaxAbs), and visualized performance with classification reports and decision trees.

Technologies: Python, Scikit-learn, Pandas, Matplotlib, Seaborn, Classification Models, Label Encoding, Evaluation Metrics

Mental Health Chatbot May-2025- GitHub-link

- Developed an intelligent conversational chatbot as **part of the DEPI Internship Program**, focusing on mental health support and empathetic AI responses.
- Engineered and trained a mental health support chatbot using LSTM with attention, enabling context-aware conversations.
- Deployed via Flask for real-time use, improving accessibility for users in need.

Technologies: Python, Flask, TensorFlow, Keras, Git, GitHub, HTML, CSS, JavaScript, Git LFS.

Sensitive Data Chat Monitor Apr-2025-GitHub-link

- Implemented a secure chat interface to prevent data leaks by detecting and blocking sensitive keywords using keyword logic and user intent control.
- Enhanced safety in internal communications through a Flask-Node.js integration.

Technologies: HTML, JavaScript, CSS, Node.js, Express.

Clustering Analysis of Wholesale Customer Data Apr-2025-GitHub-link

- Applied clustering techniques (KMeans and DBSCAN) to segment wholesale customers based on annual spending across product categories.
- Performed outlier removal and data standardization to improve model accuracy and robustness.
- Utilized the Elbow Method and Silhouette Score for optimal cluster selection.
- Visualized results using PCA for dimensionality reduction and interpretability.

Technologies: Python, Scikit-learn, Pandas, Matplotlib, Seaborn, PCA, Clustering Metrics.

NYC Taxi Trip Analysis Dashboard Mar-2025

- Built an interactive analytical dashboard using Plotly Dash to explore NYC taxi trip data, including heatmaps, regression-based fare prediction, and trip duration analysis.
- Enabled real-time data filtering and summary insights.

Technologies: Python, Plotly Dash, Pandas, Mapbox, Scikit-learn, Data Visualization, Linear Regression.

Machine Learning Models Evaluation Dec-2024 – GitHub-link

- Python-based tools to analyze and evaluate datasets using machine learning techniques (e.g., classification, regression, and clustering).
- Enhanced decision-making in selecting the best machine learning models for different tasks.

Technologies: python, ML Techniques, Scikit-learn, Pandas, Matplotlib, Seaborn.

House Price Regression: Scaling Method Impact Kaggle-link

- Conducted a comparative study on how different feature scaling techniques (Min Max Scaler, Standard Scaler, Robust Scaler, Max Absolute Scaler) affect model performance on house price prediction.
- Evaluated and visualized the performance of Linear Regression and K-Nearest Neighbors (KNN) under each scaling method using R^2 Score.
- Demonstrated the critical role of preprocessing in non-parametric models like KNN, while showing Linear Regression's stability across scalers.

Technologies: Python, Pandas, Scikit-learn, Matplotlib, Seaborn, Regression, Feature Scaling.

Prosthetic Hand project 4^{th} semester - Documentary

- Designed a smart prosthetic hand to improve upper-limb functionality for amputees.
- Used IOT and robotics to simulate natural hand movements.

Technologies: ML, IOT technology, Robotics.

Radio Labelling for Radio Stations Dec-2023 – GitHub-link

- Solution to interference between communications that arises due to interference from uncontrolled simultaneous transmissions, two channels close to each other can interfere or interact and thus cause damage to communication.

Technologies: python.

K-means implementation GitHub-link

- Implementing k-means without using the library, according to the user's choice k to determine the centroid (random, deterministic, which is dividing the data K and choosing the critical point as the centroid), calculating the Euclidean and Manhattan distance.

Technologies: python.

Reproducible Machine Learning Pipeline May-2023 - GitHub-link

- Structured a reproducible end-to-end Jupyter machine learning workflow covering data loading, exploratory analysis, preprocessing, model training, and evaluation.
- Documented environment setup and a clear migration path from notebooks to production-ready scripts.

Technologies: Python, Jupyter Notebook, Data Analysis, Data Preprocessing, Machine Learning, Model Training, Model Evaluation, Reproducible Workflows.

Training & Courses

- Generative AI Intern | DEPI – eyouth **Nov 2025 - Present**
  - *Training included:* Deep Learning, LLMs, Transformers, GANs, Diffusion Models, HuggingFace.
- AI Intern | The British University in Egypt **Feb 2025**
  - *Training included:* Computer vision, CNN models, Transfer Learning, YOLO basics.
- AI & ML from Shabab Mubtakirun & Huawei **Jan – Feb 2025**
  - *Training included:* ML pipelines, Regression, Classification, Clustering, Data preprocessing.
- Data science Intern | DEPI-AMIT **Oct 2024 – May 2025**
  - *Training included:* Data cleaning, EDA, Feature Engineering, Statistics, Model evaluation.
- Frontend Development course – Coursera
  - *Course included:* HTML, CSS, JavaScript.
- Java course – Coursera
- Machine Learning course- Coursera
  - *Training included:* KNN, SVM, Decision Tree, K-Means, model evaluation metrics.
- Python course- Coursera

Skills

- **Technical Skills:**
  - **programming language:** python, Java, JavaScript.
  - Object Oriented Programming (OOP).
  - **Web Development:** HTML, CSS, PHP, Node.js, Django, Flask.
  - **Machine Learning:** Supervised, Unsupervised, Reinforcement Learning.
  - **Deep Learning:** TensorFlow, NLP, LSTM.
  - Data Analysis & Manipulation.
  - **Data Visualization:** Matplotlib, Seaborn, Plotly, Dash.
  - **Big Data Tools:** Apache Spark.
  - Algorithms & Problem Solving.
- **Personal Skills:**
  - problem-solving Abilities.
  - Attention to Detail.
  - Effective Time management.
  - Continuous Learning.